

Vedansh Somani

Undergraduate Student

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Profile

3rd-year Robotics & AI student who builds across the stack — from ROS2-based autonomous robots and RL pipelines to full-stack mobile apps. I'm most at home where software meets the physical world.

Education

BE - Robotics and AI,

Dayananda Sagar College of Engineering

09/2023 – Present | Bengaluru, India

CBSE - Science, Sai International Residential School

2021 – 2023 | Cuttack, India

Professional Experience

STEM Trainer, Beylabee

04/2024 – 04/2025 | Bengaluru, India

Skills

Programming

- Python
- C++
- HTML
- CSS
- JavaScript
- Nodejs
- React
- Flutter
- ROS

Reinforcement Learning

- Deep Q-Learning (DQN)
- Epsilon-Greedy Exploration
- Reward Shaping
- Q-Network Design (spatial CNN)

Microcontrollers and Embedded Systems

ESP32 and Arduino

Projects

Autonomous Mobile Robot

02/2026 – 05/2026

Built a ROS2-based autonomous mobile robot featuring SLAM, autonomous navigation via Nav2 stack, and real-time obstacle avoidance using LiDAR.

Chain Reaction + RL

05/2026 – 05/2026

Recreated the classic Chain Reaction board game in Python with pygame and trained a DQN-based AI opponent from scratch using PyTorch. Built a full RL pipeline with custom reward shaping, a four-stage curriculum, and a self-aware lookahead filter — achieving a 57% win rate against a greedy policy baseline.

Chat Application

12/2025 – Present

Created a mobile chat application with encrypted login, backend server, socket connections, message database and organized UI.

Jarvis-Lite Assistant

01/2026 – Present

Created a Jarvis like assistant which listens to voice commands, and executes the ones that are permitted based on existing ruleset.

4WD-Rover

07/2025 – 08/2025

A 4WD rover built using a Cytron Microcontroller and variable inputs (Webpage hosted on server, Flysky, bluetooth, etc.)